ext3, ReiserFS, JFS and XFS. We will use the most universal choice, ext3. Create an ext3 file system on */dev/sda1*, using *mkfs.ext3*, as follows:

```
[livecd /]# mkfs.ext3 /dev/sda1
mke2fs 1.41.12 (17-May-2010)
Filesystem label=
OS type: Linux
<..omitted..>
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

This filesystem will be automatically checked every 34 mounts or
180 days, whichever comes first.  Use tune2fs -c or -i to override.
```

To create swap, we use *mkswap*, and then use *swapon* to start using the partition as a swap device:

```
[livecd /]# mkswap /dev/sda2
Setting up swapspace version 1, size = 714792 KiB
no label, UUID=7405709f-fe07-412d-b378-5c6e1aec7b71
[livecd /]# swapon /dev/sda2
```

Mount the root partition as follows:

```
[livecd /]# mount /dev/sda1 /mnt/gentoo
[livecd /mnt/gentoo]# ls
lost+found
```

We are now ready to move on to the next step—downloading and extracting stage3 and Portage.

Stage3 set-up

Select a download mirror close to you from the list at http://www.gentoo.org/main/en/mirrors2.xml and navigate to *releases/x86/current-stage3/* on that site. Download and extract stage3 as follows:

```
[livecd /]# cd /mnt/gentoo
[livecd /mnt/gentoo]# wget http://mirrors.kernel.org/gentoo/releases/x86/
current-stage3/stage3-i686-*.tar.bz2
Resolving mirrors.kernel.org...
Connecting to mirrors.kernel.org|80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 127443348 (122M) [application/x-bzip2]
Saving to: stage3-i686-*.tar.bz2

100%[==================================>] 127,443,348 1.19M/s   in 1m 50s

1.10 MB/s - stage3-i686-*.tar.bz2 saved [127443348/127443348]

[livecd /mnt/gentoo]# tar xvjpf stage3-i686-*.tar.bz2
<...omitted...>
[livecd /mnt/gentoo]# ls
bin  boot  dev  etc  home  lib  mnt  opt  proc  root  sbin
stage3-i686-20100601.tar.bz2  sys  tmp  usr  var
```

Portage set-up

Download and extract Portage as shown Figure 2.

Chroot into the extracted system

Before we move on to the next step, we need to chroot the system (make the */mnt/gentoo* directory the root for the current terminal session's shell). First, we must make sure that the network works under the *chroot* environment. Copy the DNS information from the live CD environment into the mounted partition, as follows:

```
[livecd /mnt/gentoo/usr]# cd
```

```
=====CODE=====
[livecd /mnt/gentoo]# cd usr
[livecd /mnt/gentoo/usr]# wget http://mirrors.kernel.org/gentoo/snapshots/portage-latest.tar.bz2
Resolving mirrors.kernel.org...
Connecting to mirrors.kernel.org|80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 37570038 (36M) [application/x-bzip2]
Saving to: `portage-latest.tar.bz2'

100%[===========================================================================>] 37,570,038  1.22M/s   in 30s

1.21 MB/s - `portage-latest.tar.bz2' saved [37570038/37570038]
[livecd /mnt/gentoo]# tar xvjpf portage-latest.tar.bz2
<...omitted...>
[livecd /mnt/gentoo/usr]# ls portage
app-accessibility  app-mobilephone  dev-libs      games-arcade      gnome-extra     media-libs     net-misc         sci-chemistry          sys-cluster  x11-drivers
app-admin          app-office       dev-lisp      games-board       gnustep-apps    media-plugins  net-nds          sci-electronics        sys-devel    x11-libs
app-antivirus      app-pda          dev-ml        games-emulation   gnustep-base    media-radio    net-news         sci-geosciences        sys-freebsd  x11-misc
app-arch           app-portage      dev-perl      games-engines     gnustep-libs    media-sound    net-nntp         sci-libs               sys-fs       x11-plugins
app-backup         app-shells       dev-php       games-fps         gpe-base        media-tv       net-p2p          sci-mathematics        sys-kernel   x11-proto
app-benchmarks     app-text         dev-php5      games-kids        gpe-utils       media-video    net-print        sci-misc               sys-libs     x11-terms
app-cdr            app-vim          dev-python    games-misc        header.txt      metadata       net-proxy        sci-physics            sys-power    x11-themes
app-crypt          app-xemacs       dev-ruby      games-mud         java-virtuals   net-analyzer   net-voip         sci-visualization      sys-process  x11-wm
app-dicts          dev-ada          dev-scheme    games-puzzle      kde-base        net-dialup     net-wireless     scripts                virtual      xfce-base
app-doc            dev-cpp          dev-tcltk     games-roguelike   kde-misc        net-dns        net-zope         sec-policy             www-apache   xfce-extra
app-editors        dev-db           dev-tex       games-rpg         licenses        net-firewall   perl-core        skel.ChangeLog         www-apps
app-emacs          dev-dotnet       dev-texlive   games-server      lxde-base       net-fs         profiles         skel.ebuild            www-client
app-emulation      dev-embedded     dev-tinyos    games-simulation  mail-client     net-ftp        rox-base         skel.metadata.xml      www-misc
app-forensics      dev-games        dev-util      games-sports      mail-filter     net-im         rox-extra        sys-apps               www-plugins
app-i18n           dev-haskell      dev-vcs       games-strategy    mail-mta        net-irc        sci-astronomy    sys-auth               www-servers
app-laptop         dev-java         eclass        games-util        media-fonts     net-libs       sci-biology      sys-block              x11-apps
app-misc           dev-lang         games-action  gnome-base        media-gfx       net-mail       sci-calculators  sys-boot               x11-base
[livecd /mnt/gentoo/usr]#
=====CODE=====
```

Figure 2: Portage set-up